

Lesson 7. Vector calculus II

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Divergence

Consider in Cartesian coordinates the three-dimensional vector field:

$$\vec{F}(x, y, z) = \langle P(x, y, z), Q(x, y, z), R(x, y, z) \rangle$$

The **divergence** of a vector field measures the fluid flow "out of" or "into" a given point. The **curl** indicates how much the fluid rotates or spins around a point.

The **divergence** of $\vec{F}$, denoted as $\operatorname{div} \vec{F}$ is defined as the scalar field given by

$$\begin{aligned}\operatorname{div} \vec{F} = \nabla \cdot \vec{F} &= \mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z} \cdot (P\mathbf{i} + Q\mathbf{j} + R\mathbf{k}) = \\ &= \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}.\end{aligned}$$

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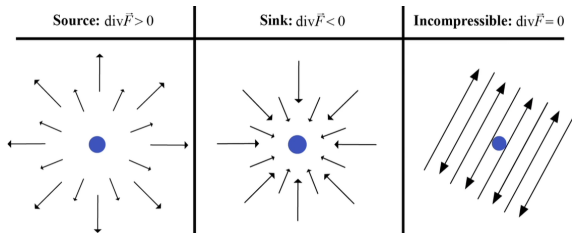
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Divergence

Properties

- If $\nabla \cdot \mathbf{F} > 0$, the vector field is a **source**, and the net rate of change of the fluid is flowing **from a point**.
- If $\nabla \cdot \mathbf{F} < 0$, the vector field is a **sink**, and the net rate of change of the fluid is flowing **toward a point**.
- If $\nabla \cdot \mathbf{F} = 0$, then $\mathbf{F}$ is said to be **incompressible** (no sources or sinks).
- If $\mathbf{F}$ is a vector field on $\mathbb{R}^3$ and P , Q , and R have continuous second-order partial derivatives, then $\operatorname{div}(\operatorname{curl} \vec{\mathbf{F}}) = \nabla \cdot (\nabla \times \vec{\mathbf{F}}) = 0$.



Divergence

Example: Consider the vector fields $\mathbf{F} = \langle -y, x \rangle$ and $\mathbf{R}(x, y) = \langle -x, -y \rangle$.

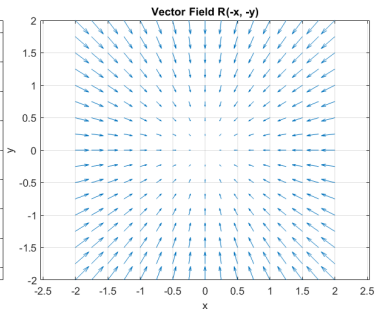
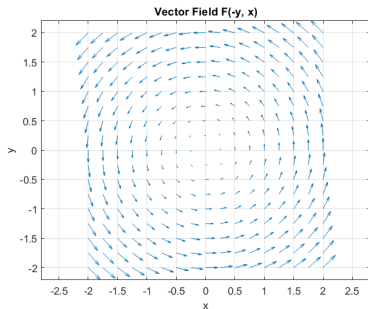
The divergence of $\mathbf{F}$ is:

$$\operatorname{div}(\langle -y, x \rangle) = \frac{\partial}{\partial x}(-y) + \frac{\partial}{\partial y}(x) = 0.$$

The divergence of $\mathbf{R}$ is:

$$\operatorname{div}(\mathbf{R}) = \frac{\partial}{\partial x}(-x) + \frac{\partial}{\partial y}(-y) = -2.$$

Divergence



$\mathbf{F} = \langle -y, x \rangle$: At any particular point, the amount of fluid flowing in is the same as the amount flowing out, so at every point, the "outflowingness" of the field is zero (incompressible vector field). Therefore, the divergence of the $\mathbf{F}$ is zero.

$\mathbf{R}(x, y) = \langle -x, -y \rangle$: For the second field, at any given point, more fluid is flowing in than is flowing out, and therefore, the "outflowingness" of the field is negative. Thus, for the $\mathbf{R}$, the divergence of this field is negative.

Curl

If $\mathbf{F}$ is a vector field on $\mathbb{R}^3$ and the partial derivatives of P , Q , and R exist, then the **curl** of $\mathbf{F}$ is the vector field on $\mathbb{R}^3$ defined by:

$$\operatorname{curl} \vec{F} = \nabla \times \mathbf{F} = \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z} \right) \vec{i} + \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x} \right) \vec{j} + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \vec{k}$$

$$\operatorname{curl} \vec{F} = \nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix} = \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z} \right) \mathbf{i} - \left(\frac{\partial R}{\partial x} - \frac{\partial P}{\partial z} \right) \mathbf{j} + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \mathbf{k}$$

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Curl

Properties

- If f is a function of three variables that has continuous second partial derivatives, then $\nabla \times (\nabla f) = \mathbf{0}$, indicating that the vector field is irrotational.
- If $\mathbf{F}$ is a vector field defined on all of $\mathbb{R}^3$ whose component functions have continuous partial derivatives and $\nabla \times \mathbf{F} = \mathbf{0}$, then $\mathbf{F}$ is a conservative vector field.
- The **curl** $\nabla \times \mathbf{F}$ indicates the direction about which the fluid rotates most rapidly, and the speed of this rotation is found by the magnitude $|\nabla \times \mathbf{F}|$.
- The **curl** follows the right-hand rule, such that a counterclockwise rotation indicates a positive curl.

Curl

Find the curl of

$$\vec{F}(P, Q, R) = \langle x^2z, e^y + xz, xyz \rangle.$$

Solution

The curl is

$$\begin{aligned} \text{curl } f &= \nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2z & e^y + xz & xyz \end{vmatrix} = \\ &= \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z} \right) \hat{i} + \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x} \right) \hat{j} + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \hat{k} = \\ &= \left(\frac{\partial}{\partial y}(xyz) - \frac{\partial}{\partial z}(e^y + xz) \right) \hat{i} + \left(\frac{\partial}{\partial z}(x^2z) - \frac{\partial}{\partial x}(xyz) \right) \hat{j} + \left(\frac{\partial}{\partial x}(e^y + xz) - \frac{\partial}{\partial y}(x^2z) \right) \hat{k} = \\ &= (xz - x) \hat{i} + (x^2 - yz) \hat{j} + z \hat{k}. \end{aligned}$$

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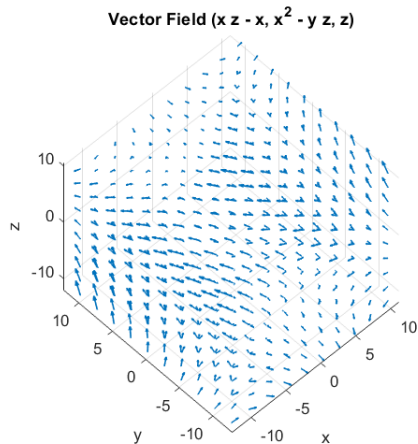
7.1. Divergence and Curl

7.2. Parametric Surface

7.3. Surface Integrals

7.4. Stokes' Theorem, Divergence Theorem

Curl



The curl and divergence operators allow us to rewrite Green's theorem in an alternative form.

Vector form of Green's Theorem using Curl

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_D (\text{curl } \vec{F}) \cdot \vec{k} dA$$

Vector form of Green's Theorem using Divergence

$$\oint_C \vec{F} \cdot \vec{n} ds = \iint_D \text{div } \vec{F}(x, y) dA$$

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How do you parameterize a surface?

This is the big question we need to answer in order to calculate **surface integrals**, **Stokes' theorem**, and the **Divergence theorem**, which are crucial for understanding vector calculus and its relationship with PDEs.

Parameterization

Parameterization

First, we need to begin with finding a parametric representation for a surface.

Why?

Because relating a surface to a subset of a plane makes things easier to calculate, and parameterization is the tool we need to accomplish this goal.

We are looking at how to find the **parametric representation for a surface** using the following coordinates:

- Cartesian;
- Polar;
- Cylindrical;
- Spherical

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Parameterization

1. Cartesian Coordinates

In Cartesian coordinates (x, y, z) , a surface can be represented as a function $z = f(x, y)$, where x and y are the parameters.

Find the parametric equations for the surface $f(x, y) = 9 - x^2 - y^2$

One way to parameterize the surface is to take x and y as parameters and write the parametric equation as $x = x, y = y$ and $z = f(x, y)$ such that the parameterizations for this paraboloid is:

$$x = x, \quad y = y, \quad z = f(x, y) = 9 - x^2 - y^2, \quad (x, y) \in \mathbb{R}^2$$

But **parameterizations are not unique**, as we can also represent this surface using polar coordinates.

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Parameterization

2. Polar Coordinates

In polar coordinates (r, θ) , used in 2D (x, y) space, we express x and y as functions of r and θ :

$$x = r \cos(\theta), \quad y = r \sin(\theta)$$

To describe a surface, combine these with $z = f(r, \theta)$ as the third coordinate.

Find the parametric equations for the surface $f(x, y) = 9 - x^2 - y^2$

If we let $x = r \cos \theta$, $y = r \sin \theta$, and $z = f(r, \theta)$ then the paraboloid has the following parametric representation

$$x = r \cos \theta, \quad y = r \sin \theta, \quad z = f(x, y) = 9 - r^2$$

$$\text{where } r > 0 \text{ and } 0 \leq \theta \leq 2\pi$$

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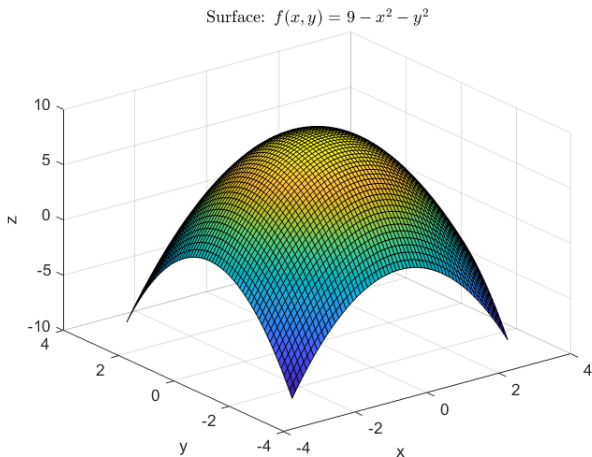
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Parameterization



Parameterization

3. Cylindrical Coordinates

In cylindrical coordinates (r, θ, z) , we use r and θ to express x and y , while z is kept as is:

$$x = r \cos(\theta), \quad y = r \sin(\theta), \quad z = z$$

This form is helpful for surfaces of revolution or surfaces with circular symmetry.

Find the parametric equations for the cylinder $y^2 + z^2 = 25$

In cylindrical coordinates

$$x = x, \quad y = 5 \sin \theta, \quad z = 5 \cos \theta$$

Thus, a parametric representation for this cylinder:

$$\vec{r}(x, \theta) = x \vec{i} + 5 \sin \theta \vec{j} + 5 \cos \theta \vec{k}$$

We will also need the restriction $0 \leq \theta \leq 2\pi$ to make sure that we don't retrace any portion of the cylinder.

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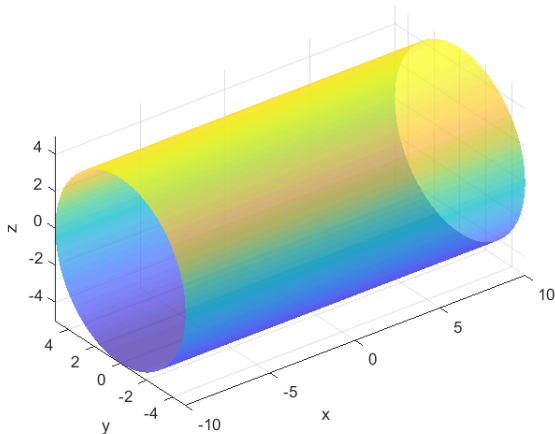
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Parameterization

Cylinder: $y^2 + z^2 = 25$



Parameterization

4. Spherical Coordinates

In spherical coordinates (ρ, θ, ϕ) , a point is represented with:

$$x = \rho \sin(\phi) \cos(\theta), \quad y = \rho \sin(\phi) \sin(\theta), \quad z = \rho \cos(\phi)$$

This system is often used for spherical surfaces or radially symmetric surfaces.

Find the parametric representation for the sphere $x^2 + y^2 + z^2 = 30$

Using the above formulas for converting Cartesian coordinates into spherical coordinates:

$$x = \sqrt{30} \sin \varphi \cos \theta, \quad y = \sqrt{30} \sin \varphi \sin \theta, \quad z = \sqrt{30} \cos \varphi$$

Therefore, the parametric representation is,

$$\vec{r}(\theta, \varphi) = \sqrt{30} \sin \varphi \cos \theta \vec{i} + \sqrt{30} \sin \varphi \sin \theta \vec{j} + \sqrt{30} \cos \varphi \vec{k}$$

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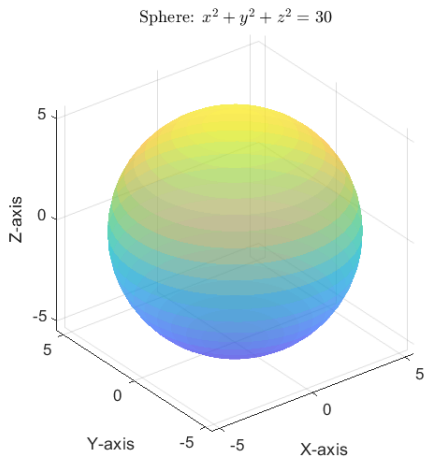
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Parameterization



Tangent Planes

Finding tangent planes to parametric surfaces

Consider a parametric surface S that is traced out by a vector function:

$$\vec{r}(u, v) = \langle x(u, v), y(u, v), z(u, v) \rangle$$

at a point P_0 with position vector $\vec{r}(u_0, v_0)$

All we have to do is find a point through the plane and the normal vector:

We already know the point, P_0 , so we just need a normal vector to this plane. We can calculate our normal vector to the tangent plane by using the cross product of our partial derivatives of $\vec{r}$ with respect to u and v at point P_0 .

$$\vec{n} = \vec{r}_u(u_0, v_0) \times \vec{r}_v(u_0, v_0)$$

It is important to know that if $\vec{r}_u \times \vec{r}_v \neq \vec{0}$, then the surface S is called **smooth** and has no corners at the point $\vec{r}(u_0, v_0)$.

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Tangent Planes

Find the equation of the tangent plane to the surface given by

$$\vec{r}(u, v) = u\vec{i} + 2v^2\vec{j} + (u^2 + v)\vec{k}$$

at the point $(2, 2, 3)$.

Solution:

The parameterization is given, so we need to take the first order partial derivatives with respect to u and v :

$$\vec{r}_u(u, v) = \vec{i} + 2u\vec{k}$$

$$\vec{r}_v(u, v) = 4v\vec{j} + \vec{k}$$

Next, we will calculate the normal vector to the tangent plane:

$$\vec{n} = \vec{r}_u \times \vec{r}_v = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 0 & 2u \\ 0 & 4v & 1 \end{vmatrix} = -8uv\vec{i} - \vec{j} + 4v\vec{k}$$

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Tangent Planes

Find the equation of the tangent plane to the surface given by

$$\vec{r}(u, v) = u\vec{i} + 2v^2\vec{j} + (u^2 + v)\vec{k}$$

at the point $(2, 2, 3)$.

Solution:

The parameterization is given, so we need to take the first order partial derivatives with respect to u and v :

$$\vec{r}_u(u, v) = \vec{i} + 2u\vec{k}$$

$$\vec{r}_v(u, v) = 4v\vec{j} + \vec{k}$$

Next, we will calculate the normal vector to the tangent plane:

$$\vec{n} = \vec{r}_u \times \vec{r}_v = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 0 & 2u \\ 0 & 4v & 1 \end{vmatrix} = -8uv\vec{i} - \vec{j} + 4v\vec{k}$$

Tangent Planes

Now, we will determine the value of u and v that will give us the point P_0 :

$$2 = u \quad \Rightarrow \quad u = 2$$

$$2 = 2v^2 \quad \Rightarrow \quad v = \pm 1$$

$$3 = u^2 + v \quad \Rightarrow \quad v = -1$$

So we now know that we'll be at the point when $u = 2$ and $v = -1$. At this point the normal vector is,

$$\vec{n} = 16\vec{i} - \vec{j} - 4\vec{k}$$

The tangent plane is then,

$$16(x - 2) - (y - 2) - 4(z - 3) = 0$$

$$16x - y - 4z = 18$$

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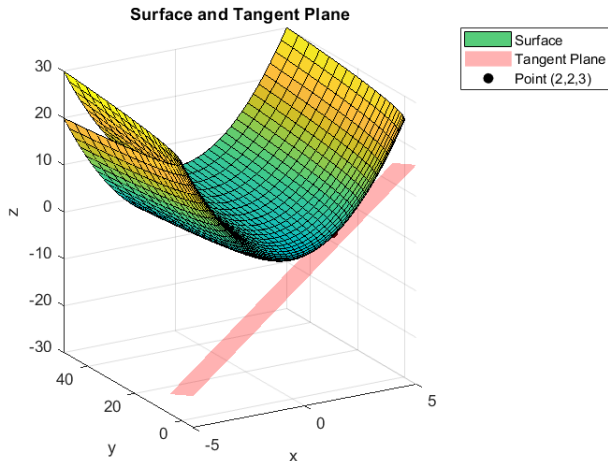
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Tangent Plan



Surface Area Of Parametric Surfaces

We can extend our parameterization skills to find the surface area of parametric surfaces using double integrals.

If a smooth parametric surface S is given by the equation:

$$\vec{r}(u, v) = \langle x(u, v), y(u, v), z(u, v) \rangle$$

where $(u, v) \in D$ and S is covered just once as (u, v) ranges throughout the parameter domain D , then the surface area of S is:

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Surface Area Of Parametric Surfaces

Proof

If we take the function $z = f(x, y)$ and parameterize it so that $\vec{r}(x, y, z) = \langle x, y, f(x, y) \rangle$ by letting $X = x, y = y$ and $z = f(x, y)$.

Then,

$$\vec{r}_x = \langle 1, 0, f_x \rangle \text{ and } \vec{r}_y = \langle 0, 1, f_y \rangle$$

And

$$\vec{r}_x \times \vec{r}_y = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 0 & f_x \\ 0 & 1 & f_y \end{vmatrix} = \langle -f_x, -f_y, 1 \rangle$$

Thus,

$$\|\vec{r}_x \times \vec{r}_y\| = \sqrt{(-f_x)^2 + (-f_y)^2 + 1^2}$$

Which is exactly the same formula we learned about in our previous lesson on surface area of the graph of a function $z = f(x, y)$:

$$A(S) = \iint_D \sqrt{1 + \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2} dA$$

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Surface Integral

What's the difference between line integrals and surface integrals?

A **line integral** evaluates a multivariable function along a line, whereas a **surface integral** calculates a multivariable function over a surface.

And just as line integrals has two forms for either scalar functions or vector fields, **surface integrals also have two forms:**

- Surface integrals of scalar functions (scalar field);
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Surface integrals of scalar field

We'll begin with the surface integral of a scalar field

- We have some surface in three-dimensions. The surface itself is a two-dimensional object.
- We have a scalar field defined on the surface.
- The surface integral of the scalar field will be obtained by a Riemann sum, where we break the surface into small surface elements dS , we multiply each element by the average value of the scalar field, say $f(\mathbf{r})$ on that surface element, and then we sum over all of those surface elements.

The surface integral of a function $f(x, y, z)$ over a surface S is written $\iint_S f(x, y, z) dS$, where dS stands for the infinitesimal amount of surface area. Thus, the surface integral of a function can be written as:

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where D is the projection onto the xy -plane.

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Surface integrals of scalar field

How do we actually compute such an integral?

- ➊ Using formula (1). Note as well that there are similar formulas for surfaces given by $y = g(x, z)$ (with D in the xz -plane) and $x = g(y, z)$ (with D in the yz -plane).
- ➋ Using formula (2)

Let S be a smooth surface described parametrically by $\vec{r}(u, v)$ where $(u, v) \in D$, and suppose the domain of $f(x, y, z)$ includes S . Then the surface integral of f over S with respect to surface area is:

$$\iint_S f(x, y, z) dS = \iint_D f(\vec{r}(u, v)) \|\vec{r}_u \times \vec{r}_v\| dA \quad (2)$$

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Surface integrals of scalar field

Example 1

Evaluate

$$\iint_S 6xy \, dS$$

where S is the portion of the plane $x + y + z = 1$ that lies in the 1st octant and is in front of the yz -plane.

Solution:

Since we are looking for the portion of the plane that lies in front of the yz -plane we are going to need to write the equation of the surface in the form $x = g(y, z)$:

$$x = 1 - y - z$$

Now, the surface integral of the given scalar field is of the form:

$$\iint_S f(x, y, z) \, dS = \iint_D f(g(y, z), y, z) \sqrt{1 + \left(\frac{\partial g}{\partial y}\right)^2 + \left(\frac{\partial g}{\partial z}\right)^2} \, dA$$

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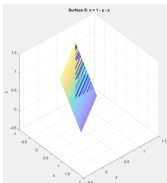
Surface integrals of scalar field

Example 1

$$\iint_S 6xy \, dS = \iint_D 6(1-y-z)y \sqrt{1+(-1)^2+(-1)^2} \, dA =$$

At this point we've got a double integral to do:

$$= \sqrt{3} \iint_D 6(y - y^2 - zy) \, dA$$



We arrived at the equation of the hypotenuse by setting x equal to zero in the equation of the plane and solving for z . Here are the ranges for y and z :

$$0 \leq y \leq 1, \quad 0 \leq z \leq 1 - y$$

Surface integrals of scalar field

Example 1

$$\begin{aligned}\iint_S 6xy \, dS &= \sqrt{3} \iint_D 6(y - y^2 - zy) \, dA \\&= 6\sqrt{3} \int_0^1 \int_0^{1-y} (y - y^2 - zy) \, dz \, dy \\&= 6\sqrt{3} \int_0^1 \left(yz - zy^2 - \frac{1}{2}z^2y \right) \Big|_0^{1-y} dy \\&= 6\sqrt{3} \int_0^1 \left(\frac{1}{2}y - y^2 + \frac{1}{2}y^3 \right) dy \\&= 6\sqrt{3} \left(\frac{1}{4}y^2 - \frac{1}{3}y^3 + \frac{1}{8}y^4 \right) \Big|_0^1 = \frac{\sqrt{3}}{4}\end{aligned}$$

Surface integrals of scalar field

Example 2

Compute the surface integral of $f(x, y, z) = yz$ over S with respect to surface area if S is described parametrically as $\vec{r}(u, v) = \langle u^2, u \sin v, u \cos v \rangle$ with $u \in [0, 1], v \in [0, \frac{\pi}{2}]$.

Solution:

First, we need to find the first order partials:

$$\vec{r}_u = \langle 2u, \sin v, \cos v \rangle \text{ and } \vec{r}_v = \langle 0, u \cos v, -u \sin v \rangle$$

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Next, we will compute the cross product $\vec{r}_u \times \vec{r}_v$:

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Lastly, we substitute everything into our surface integral formula and solve:

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Surface integrals of scalar field

Example 2

$$\begin{aligned}\iint_S (yz) dS &= \int_0^1 \int_0^{\frac{\pi}{2}} (u^2 \sin v \cos v) (u \sqrt{1 + 4u^2}) dv du \\ &= \frac{1}{64} \left(\frac{50\sqrt{5} - 2}{5} + \frac{10\sqrt{5} - 2}{3} \right) \approx 0.449\end{aligned}$$

Surface integrals of vector field

Surface Integral Orientation

For a surface $z = g(x, y)$ the orientation by the unit normal vector is:

$$\vec{n} = \frac{\left\langle -\frac{\partial g}{\partial x}, -\frac{\partial g}{\partial y}, 1 \right\rangle}{\sqrt{1 + \left(\frac{\partial g}{\partial x}\right)^2 + \left(\frac{\partial g}{\partial y}\right)^2}} \quad (1)$$

Surface Integral Orientation

And a parametric smooth surface $S : \{\vec{r}(u, v) : (u, v) \in D\}$ the orientation of the unit normal vector is:

$$\vec{n} = \frac{\vec{r}_u \times \vec{r}_v}{\|\vec{r}_u \times \vec{r}_v\|} \quad (2)$$

Surface integrals of vector field

Surface Integral Orientation

For a surface $z = g(x, y)$ the orientation by the unit normal vector is:

$$\vec{n} = \frac{\left\langle -\frac{\partial g}{\partial x}, -\frac{\partial g}{\partial y}, 1 \right\rangle}{\sqrt{1 + \left(\frac{\partial g}{\partial x}\right)^2 + \left(\frac{\partial g}{\partial y}\right)^2}} \quad (1)$$

Surface Integral Orientation

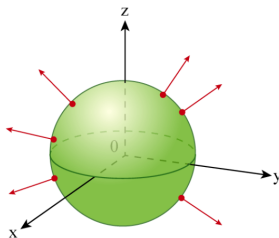
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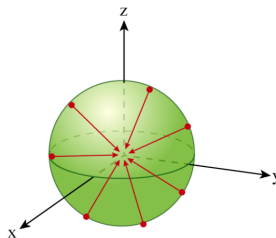
Surface integrals of vector field

Surface Integral Orientation

So, for a surface that is the boundary of a solid E (i.e., closed surface), the convention is that the positive orientation is the one for which the normal vectors point outward from E . Inward-pointing normal vectors give the negative orientation.



Positive Orientation



Negative Orientation

Surface integrals of vector field

If $\vec{F}$ is a continuous vector field defined on an oriented surface S with unit normal vector $\vec{n}$, then the surface integral of $\vec{F}$ over S is:

$$\iint_S \vec{F} \cdot d\vec{S} = \iint_S \vec{F} \cdot \vec{n} dS \quad (3)$$

This integral is also called the flux of $\vec{F}$ across S .

Why is the surface integral the flux?

Because the surface integral of a vector field represents the amount of fluid flowing through the surface per unit of time:

$$\text{Flux} = (\text{flow rate})(\text{segment length})$$

So, if a surface integral measures the total rate of flow, then we can simply call it a flux integral, because that's exactly what we are calculating - the flux!

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Surface integrals of vector field

Flux Integral – Alternate Form Suppose S is a smooth, two-sided surface given by $z = f(x, y, z)$ where (x, y) is in D and let $\vec{n}$ denote the upward unit normal vector on S . And if f has continuous first partial derivatives and $\vec{F} = \langle P, Q, R \rangle$ is a continuous vector field, then the flux of $\vec{F}$ across S is given by:

$$\iint_S \vec{F} \cdot d\vec{S} = \iint_D \left(-P \frac{\partial f}{\partial x} - Q \frac{\partial f}{\partial y} + R \right) dA \quad (4)$$

OR

$$\iint_S \vec{F} \cdot d\vec{S} = \iint_D \vec{F}(\vec{r}(u, v)) \cdot (\vec{r}_u \times \vec{r}_v) dA \quad (5)$$

Surface integrals of vector field

Example 1: Calculate the surface integral

$$\iint_S \mathbf{F} \cdot \mathbf{N} \, dS,$$

where $\mathbf{F} = \langle -y, x, 0 \rangle$ and S is the surface with parameterization

$$\mathbf{r}(u, v) = \langle u, v^2 - u, u + v \rangle, \quad 0 \leq u \leq 3, \quad 0 \leq v \leq 4.$$

Solution:

The tangent vectors are $\mathbf{r}_u = \langle 1, -1, 1 \rangle$ and $\mathbf{r}_v = \langle 0, 2v, 1 \rangle$. Therefore,

$$\mathbf{r}_u \times \mathbf{r}_v = \langle -1 - 2v, -1, 2v \rangle.$$

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Surface integrals of vector field

Solution:

By Equation (5),

$$\begin{aligned}\iint_S \mathbf{F} \cdot d\mathbf{S} &= \int_0^4 \int_0^3 \mathbf{F}(\mathbf{r}(u, v)) \cdot (\mathbf{t}_u \times \mathbf{t}_v) \, du \, dv = \\ &= \int_0^4 \int_0^3 \langle u - v^2, u, 0 \rangle \cdot \langle -1 - 2v, -1, 2v \rangle \, du \, dv = \\ &= \int_0^4 \int_0^3 [(u - v^2)(-1 - 2v) - u] \, du \, dv = \int_0^4 \int_0^3 (2v^3 + v^2 - 2uv - 2u) \, du \, dv = \\ &= \int_0^4 [2v^3u + v^2u - vu^2 - u^2]_0^3 \, dv = \\ &= \int_0^4 (6v^3 + 3v^2 - 9v - 9) \, dv = \left[\frac{3v^4}{2} + v^3 - \frac{9v^2}{2} - 9v \right]_0^4 = 340.\end{aligned}$$

Therefore, the flux of $\mathbf{F}$ across S is 340.

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Surface integrals of vector field

Example 2: Calculate the surface integral

$$\iint_S \mathbf{F} \cdot \mathbf{N} dS,$$

where $\mathbf{F} = \langle 0, -z, y \rangle$ and S is the portion of the unit sphere in the first octant with outward orientation.

Solution:

Step 1: Parametrize the Surface

The unit sphere can be parametrized using spherical coordinates. For the first octant, we have:

$$x = \rho \sin \phi \cos \theta, y = \rho \sin \phi \sin \theta, z = \rho \cos \phi,$$

where: $\rho = 1$ (since it's a unit sphere), ϕ ranges from 0 to $\frac{\pi}{2}$, θ ranges from 0 to $\frac{\pi}{2}$. Thus, the parametrization simplifies to:

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Surface integrals of vector field

Solution:

Step 2: Find the Surface Element

The surface area element dS on the sphere in spherical coordinates is given by:

$$dS = \rho^2 \sin \phi \, d\phi \, d\theta.$$

For the unit sphere ($\rho = 1$):

$$dS = \sin \phi \, d\phi \, d\theta.$$

Step 3: Calculate the Normal Vector

For an outward normal vector on the surface of the unit sphere, we have:

$$\mathbf{N} = \langle x, y, z \rangle = \langle \sin \phi \cos \theta, \sin \phi \sin \theta, \cos \phi \rangle.$$

Surface integrals of vector field

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Surface integrals of vector field

Solution:

Step 4: Evaluate $\mathbf{F} \cdot \mathbf{N}$

Now we evaluate the dot product:

$$\mathbf{F} \cdot \mathbf{N} = \langle 0, -z, y \rangle \cdot \langle \sin \phi \cos \theta, \sin \phi \sin \theta, \cos \phi \rangle.$$

Substituting for z and y :

$$\begin{aligned} &= 0 \cdot \sin \phi \cos \theta + (-\cos \phi) \cdot (\sin \phi \sin \theta) + (\sin \phi \sin \theta) \cdot \cos \phi \\ &= 0 + (-\cos \phi \sin \phi \sin \theta) + (\sin \phi \sin \theta \cos \phi) \\ &= -\cos \phi \sin \phi \sin \theta + \sin \phi \sin \theta \cos \phi = 0. \end{aligned}$$

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Surface integrals of vector field

Solution:

Step 5: Setting Up the Integral

Since the integrand $\mathbf{F} \cdot \mathbf{N}$ is zero, the integral becomes:

$$\iint_S \mathbf{F} \cdot \mathbf{N} \, dS = \iint_S 0 \, dS = 0.$$

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- 4 7.4. Stokes' Theorem. Divergence Theorem

Stokes' Theorem

Let S be an oriented smooth surface that is bounded by a simple, closed, smooth boundary curve C with positive orientation. Also let $\vec{F}$ be a vector field then,

$$\int_C \vec{F} \cdot d\vec{r} = \iint_S \text{curl} \vec{F} \cdot d\vec{S}$$

Notes

- In this theorem note that the surface S can be **any surface** so long as its boundary curve is given by C .
- Stokes' Theorem essentially says that the total amount of rotation (curl) flowing through a surface is equal to the total rotation that you would observe by walking around the boundary of that surface.

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Stokes' Theorem

Verifying Stokes' Theorem for a Specific Case

Verify that Stokes' theorem is true for the vector field $\vec{F}(x, y, z) = \langle -z, x, 0 \rangle$ and the surface S , where S is the hemisphere oriented outward, with parameterization

$$\vec{r}(\phi, \theta) = \langle \sin \phi \cos \theta, \sin \phi \sin \theta, \cos \phi \rangle,$$

where $0 \leq \theta \leq \pi$ and $0 \leq \phi \leq \pi$.

Solution 1 (Using Stokes' Theorem)

Let C be the boundary of S . Note that C is a circle of radius 1, centered at the origin, sitting in the plane $y = 0$. This circle has the parameterization $\langle \cos t, \sin t, 0 \rangle$, where $0 \leq t \leq 2\pi$. The equation for scalar surface integrals is

$$\int_C \vec{F} \cdot d\vec{r} = \int_0^{2\pi} \langle -0, \cos t, 0 \rangle \cdot \langle -\sin t, \cos t, 0 \rangle dt = \int_0^{2\pi} \cos^2 t dt = \pi.$$

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Stokes' Theorem

Solution 2 (Without usage of Stokes' Theorem)

By the equation for vector line integrals,

$$\begin{aligned}\iint_S \operatorname{curl} \vec{F} \cdot d\vec{S} &= \iint_D \operatorname{curl} \vec{F}(\vec{r}(\varphi, \theta)) \cdot (\vec{r}_\varphi \times \vec{r}_\theta) dA = \\ &= \iint_D \langle 0, -1, 1 \rangle \cdot \langle \cos \theta \sin^2 \varphi, \sin \theta \sin^2 \varphi, \sin \varphi \cos \varphi \rangle dA = \\ &= \int_0^\pi \int_0^\pi (\sin \varphi \cos \varphi - \sin \theta \sin^2 \varphi) d\varphi d\theta = \frac{\pi}{2} \int_0^\pi \sin \theta d\theta = \pi.\end{aligned}$$

Divergence Theorem

Let E be a simple solid region and S is the boundary surface of E with positive orientation. Let $\vec{F}$ be a vector field whose components have continuous first order partial derivatives. Then,

$$\iint_S \vec{F} \cdot d\vec{S} = \iiint_E \operatorname{div} \vec{F} \, dV$$

Notes

- The Divergence Theorem (or Gauss's theorem) provides a connection between the flow of a vector field through the surface surrounding a solid region and the behavior of the vector field within the region itself.
- The theorem says that the total outward flow of a vector field $\vec{F}$ through a closed surface S (which surrounds a region E) can be found by instead calculating the divergence of $\vec{F}$ throughout the volume E .

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Divergence Theorem

Example: Applying the Divergence Theorem

Calculate the surface integral

$$\iint_S \vec{F} \cdot d\vec{S},$$

where S is the surface of the cylinder $x^2 + y^2 = 1$ with $0 \leq z \leq 2$, including the circular top and bottom, and

$$\vec{F} = \langle x^3 + yz, y^3 - \sin(xz), z - x - y \rangle.$$

Solution:

Applying the divergence theorem let's calculate this surface integral as the single triple integral:

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Divergence Theorem

Thus,

$$\begin{aligned}\iint_S \vec{F} \cdot d\vec{S} &= \iiint_E \operatorname{div} \vec{F} dV, \\ &= \iiint_E (x^2 + y^2 + 1) dV \\ &= \int_0^{2\pi} \int_0^1 \int_0^2 (r^2 + 1) r dz dr d\theta \\ &= \frac{3}{2} \int_0^{2\pi} d\theta \\ &= 3\pi.\end{aligned}$$

In the end of the lesson let's see how the knowledge of the Divergence Theorem and Stokes' Theorem can be used to convert the what's called the **integral form of Maxwell's equations** into their **differential form**.

Maxwell's equations in SI units and in integral form are given by

$$\oint_S \vec{E} \cdot d\vec{S} = \frac{q_{\text{enc}}}{\epsilon_0} \quad (\text{Gauss's law for electric fields})$$

$$\oint_S \vec{B} \cdot d\vec{S} = 0 \quad (\text{Gauss's law for magnetic fields})$$

$$\oint_C \vec{E} \cdot d\vec{r} = -\frac{d}{dt} \int_S \vec{B} \cdot d\vec{S} \quad (\text{Faraday's law})$$

$$\oint_C \vec{B} \cdot d\vec{r} = \mu_0 \left(I_{\text{enc}} + \epsilon_0 \frac{d}{dt} \int_S \vec{E} \cdot d\vec{S} \right) \quad (\text{Ampere-Maxwell law})$$

where $\vec{E}$ and $\vec{B}$ are the electric and magnetic fields, q_{enc} is the charge enclosed by the bounding surface S , and I_{enc} is the current through the bounding surface. The dimensional constants ϵ_0 and μ_0 are called the permittivity and permeability of free space.

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The transformation from integral to differential form is a straightforward application of both the divergence and Stokes' theorem:

The charge q_{enc} in the volume V and the current I_{enc} through the surface S are related to the charge density ρ and the current density $\vec{J}$ by

$$q_{\text{enc}} = \int_V \rho dV, \quad I_{\text{enc}} = \int_S \vec{J} \cdot d\vec{S}.$$

We apply the divergence theorem to the surface integrals and Stokes' theorem to the line integrals, replace q_{enc} and I_{enc} by integrals over ρ and $\vec{J}$, and combine the results to obtain

$$\int_V \left(\nabla \cdot \vec{E} - \frac{\rho}{\epsilon_0} \right) dV = 0,$$

$$\int_S \left(\nabla \times \vec{E} + \frac{\partial \vec{B}}{\partial t} \right) \cdot d\vec{S} = 0,$$

$$\int_V (\nabla \cdot \vec{B}) dV = 0,$$

$$\int_S \left(\nabla \times \vec{B} - \mu_0 \left(\vec{J} + \epsilon_0 \frac{\partial \vec{E}}{\partial t} \right) \right) \cdot d\vec{S} = 0.$$

Since the integration volumes and surfaces are of arbitrary size and shape, the integrands must vanish, and we obtain the differential forms for Maxwell's equations given by

$$\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0},$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t},$$

$$\nabla \cdot \vec{B} = 0,$$

$$\nabla \times \vec{B} = \mu_0 \left(\vec{J} + \epsilon_0 \frac{\partial \vec{E}}{\partial t} \right).$$